

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a20804342426d80fd.jsonl`  
- SHA-256: `0fd5371d1d7089aa9288aa30484077e30f0819eae354f416b7831e1aac944b5c`  
- Source modified: `2026-06-16T03:20:24+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:19:16.229Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** https://www.ecva.net/papers/eccv_2024/papers_ECCV/papers/06348.pdf

****Title:**** Two-Stage Active Learning for Efficient Temporal Action Segmentation (Su et al., ECCV 2024)

****Found via:**** Small-data / weak / timestamp-supervised TAL + active learning search

Task

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2. assistant (2026-06-16T03:19:18.562Z)

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****I cannot extract the paper's core findings**** without a readable text version (plain text, uncompressed PDF, or similar).

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1. ****A text-based PDF**** or its extracted text content
2. ****The paper's abstract and results section in plain text****
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Once you supply readable content, I can reliably extract:

- The two-stage active learning pipeline design
- Specific labeling budget reductions (% annotations saved)
- Accuracy metrics (F1, edit score, frame accuracy numbers)
- Dataset names and exact quantitative results
- Sampling criteria used (uncertainty, diversity, etc.)

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[Binary content (application/pdf, 772.6KB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579968447-hsu6x9.pdf]

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Two-Stage Active Learning for Efficient

Temporal Action Segmentation

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Abstract. Training a temporal action segmentation (TAS) model on long and untrimmed videos requires gathering framewise video annotations, which is very costly. We propose a two-stage active learning framework to efficiently learn a TAS model using only a small amount of video annotations. Our framework consists of three components that work together in each active learning iteration. 1) Using current labeled frames, we learn a TAS model and action prototypes using a novel contrastive learning method. Leveraging prototypes not only enhances the model performance, but also increases the computational efficiency of both video and frame selection for labeling, which are the next components of our framework. 2) Using the currently learned TAS model and action prototypes, we select informative unlabeled videos for annotation. To do so, we find unlabeled videos that have low alignment scores to learned action prototype sequences in labeled videos. 3) To annotate a small subset of informative frames in each selected unlabeled video, we propose a video-aligned summary selection method and an efficient greedy search algorithm. By evaluation on four benchmark datasets (50Salads, GTEA, Breakfast, CrossTask), we show that our method significantly reduces the annotation costs, while consistently surpassing baselines over active learning iterations. Our method achieves comparable or better performance than other weakly supervised methods while using a small amount of labeled frames. We further extend our framework to a semi-supervised active learning setting. To the best of our knowledge, this is the first work studying active learning for TAS.

Keywords: Action segmentation · Video alignment · Active learning

1 Introduction

Automatic understanding of human activities in long and uncured videos is

crucial for many applications, such as healthcare, robotics, security and assistive technologies [22,60,100]. This has motivated many recent works on temporal action segmentation (TAS) [17,33,46,47,65,75,100,103], whose goal is to predict a label for each frame, hence partition a long video into non-overlapping segments. Learning a TAS model, in principle, requires dense framewise annotations for many videos [17]. This is extremely costly given that untrimmed videos are often long, hence, annotators need to watch hours or days of video footage and

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0.04%totalannotatedframes

0.08%totalannotatedframes

0.12%totalannotatedframes

0.16%totalannotatedframes

Ground Truth

Active quality visual'rgb-04-1'

0.04%totalannotatedframes

0.08%totalannotatedframes

0.12%totalannotatedframes

0.16%totalannotatedframes

Ground Truth

Active quality visual'rgb

Fig. 1: Visualization of ours (left) and split-random baseline (right) model performance improvement by iteratively labeling more frames. This example corresponds to video 'rgb-04-1' in 50Salads. Note split-random baseline missed at least one large segment (blue one in middle right).

specify boundaries of different actions in every video. To reduce the annotation cost for learning, some of the existing works on TAS have focused on semi-supervised learning by using a small number of fully-annotated videos and many unannotated videos [18,104,117]. Yet, they still require fully labeling multiple long videos and it is not clear how to select the best videos for annotation and training. On the other hand, weakly-supervised approaches use weak annotations (e.g., ordered or unordered list of actions) in all training videos [13,14,21,44,55,63,73,74,89,90,99,106,123]. However, the performance of both categories of methods is significantly lower than the fully-supervised techniques. While time-stamp annotation [8,50,67,69,85] can achieve performance close to the fully-supervised setting. However, while a small number of annotated frames will be provided for training, they have been carefully selected in a very costly process of annotators watching every entire video and selecting frames that correspond to distinct actions in each video. In fact, a key limitation of such methods is the lack of effective video and frame selection for labeling to significantly reduce the annotation time and guide the labeling process efficiently.

Another approach to reduce the annotation cost is the conventional active learning (AL) setting, which iteratively selects unlabeled samples for assigning labels based on some utility functions [88]. However, almost all existing AL works have focused on sample classification, where each data sample has a single label. This is different from TAS, where each video sample has multiple action labels and frame labels depend on each other. Therefore, we need to address both video selection and frame selection for labeling. Recently, [86,87] proposed a hybrid active learning approach for action detection, which is to specify spatiotemporal regions of an action in a video. Despite its success, their models only handle trimmed video clips, where each clip contains only one action and lasts a few seconds. On the other hand, in TAS, videos are long and have multiple actions, thus leveraging action temporal dependencies is crucial. Also videos often contain background (irrelevant and uninteresting actions not present in the pre-specified dictionary of actions), which makes learning challenging.

Paper Contributions. We develop a two-stage active learning framework to learn temporal action segmentation (TAS) model in long and untrimmed videos using small annotations and with relatively small performance drop compared

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Two-Stage Active Learning for Efficient Temporal Action Segmentation 3

to the fully-supervised TAS. To the best of our knowledge this is the first work addressing active learning for TAS. In our two-stage framework, we first select informative unlabeled videos (inter-video selection) followed by selecting and annotating informative frames within them (intra-video selection). For inter-video selection, we develop a method based on dynamic time warping [27] to

select unlabeled videos with diverse and distinct action orderings with respect to the action orderings in the labeled videos. This leads to increased diversity among labeled videos and subsequently to more efficient TAS learning compared to a random video selection strategy. On the other hand, instead of labeling the entire selected videos, we propose a new alignment-based video summarization method to select and annotate a sequence of a small number of frames from each selected video that aligns well with the video, hence, can capture distinct actions. We repeat this iterative annotation process until the total budget or target performance met. This strategy not only improves the AL performance compared to other frame selection criteria (see Fig. 1), but also significantly reduces annotation effort. Our method also consists of action prototypes that will be learned in conjunction with TAS using a novel regularized contrastive action prototype loss. Action prototypes allow us to perform alignment more efficiently to perform video selection. By extensive evaluation on four benchmark datasets (GTEA [34], Breakfast [54], CrossTask [135], 50Salads [107]), we show that our framework significantly boosts performance over baselines. Moreover, we expand our framework to include a semi-supervised AL setting, and we achieve comparable or better performance than other weakly supervised methods with only a minimal number of labeled frames.

2 Related Works

Temporal Action Segmentation. Fully-supervised TAS methods have used dense framewise annotations for training, including recurrent networks [20, 46, 89, 102], temporal convolution nets [24, 33, 40, 43, 58, 59, 61, 65, 66, 84, 103, 105, 113, 114, 130], graph neural networks [46, 128], transformers [4, 6, 23, 25, 70, 94, 109, 110, 115, 125, 134], combination of transformers and CNNs [75] and diffusion models [68].

Multi-stage models [4, 33, 65, 75, 75, 84, 100, 102, 113, 119, 125] have been particularly effective because of their ability to capture and refine temporal context and reduce oversegmentation [17]. Notable examples include MS-TCN [33] and its variants, employing temporal convolution networks, and ASFormer [125], which use Transformers [111]. Semi-supervised approaches provide framewise labels for a subset of videos [18, 104, 117], while weakly-supervised methods rely weak annotation, examples are action set or transcript methods [13, 14, 21, 44, 55, 63, 73, 74, 89, 90, 99, 106, 123], single-frame labels [8, 50, 67, 69, 85], activity labels [18, 19], and from other sources (e.g., narrations, subtitles) [37, 97]. To further reduce reliance on annotation, unsupervised models [1, 5, 26, 29, 31, 41, 56, 93, 95, 95, 98, 101, 112, 118] often use videos from same activity at a time or use self-supervision from large-scale datasets (e.g. Howto100M [81]), treating

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TAS as a downstream task. We argue that reducing annotation effort for TAS is important, however, instead of completely removing it, we propose an active learning framework to efficiently label only a subset of frames for a small number of videos to train TAS.

Sequence Alignment. Dynamic Time Warping (DTW) [82, 92] and its variants, such as dropDTW [27], assume ordered matching between sequences and have achieved success in various applications [10, 13, 14, 16, 16, 27, 29, 42, 101, 101, 124, 132]. To handle alignment of sequences that do not follow ordered matchings, other methods have been proposed, such as (restricted) edit distance [38, 51, 62, 99], order-preserving Wasserstein distance [108], temporal cycle consistency [30]. Our video-aligned summary selection method for intra-video selection is an extension of DTW for simultaneous summary selection and alignment.

Active Learning. The goal of active learning is to minimize the amount of data annotation by iteratively selecting samples, labeling them and training a model [88]. While query-based [3, 77, 79, 133] and stream-based [32, 53, 83] methods have been proposed, the current mainstream favors pool-based frameworks. These pool-based algorithms select a subset of samples from unlabeled data for annotation, mainly guided by two key criteria: uncertainty [7, 36, 39, 45, 49, 71, 71, 72, 76, 116, 126, 127, 131] and diversity [2, 78, 96]. Uncertainty-based methods select difficult samples about whose label the current model is most uncertain. Diversity-based methods ensure the chosen samples for labeling capture the distribution of the original data. Recent efforts in active learning have been expanded into video domain [48, 52, 64, 86, 87, 120, 136] and different settings, e.g., active fine-tuning [122] active self-supervised learning [9], active training [57] and active federated learning [11].

In the paper, we focus on untrimmed videos and explore two-stage active learning for action segmentation by capturing temporal diversity at various levels.

3 Active Temporal Action Segmentation Learning

Problem Setting. Assume we have a set of N unlabeled videos with A action classes, including background (i.e., irrelevant actions outside the pre-defined list of main action classes). For more efficient processing, we divide each video into clips, where each clip consists of a small number (e.g., 10 or 32) of consecutive frames. Each video i has T_i clips and consists of pre-extracted clip features,

$$X_i = \begin{bmatrix} x_{i1} & x_{i2} & \dots & x_{iT_i} \end{bmatrix}$$

, where x_{ij} denotes the feature of clip j in video i . We denote the set of all videos by $X = \{X_1, X_2, \dots, X_N\}$. Our two-stage active learning framework for TAS consists of c iterations, where in each iteration, we select $m \leq N$ unlabeled videos and partially annotate $\frac{1}{c}$ fraction of their clips. We denote the set of labeled video clips in iteration t by $X(t)$ and the set of remaining unlabeled video clips by $X(t)$.

It is important to note that our framework can work with any existing TAS architecture.

Overview of Proposed Framework. Our two-stage active TAS learning consists of the following components (see Fig. 2):

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Two-Stage Active Learning for Efficient Temporal Action Segmentation

Selected Summaries

TAS Backbone

Partially Labeled Videos

Prototype Learning TAS Classifier

Unlabeled Videos

TAS Backbone

? VPA

VAS?

Selected Unlabeled Videos

Summary Annotation

TAS Learning Inter-Video Selection Intra-Video Selection

Fig. 2: Overview of our two-stage AL framework. Left: we first train a TAS model and construct action prototype sequences from partially labeled videos. Middle: Informative unlabeled videos are identified through our inter-video selection method (VPA). Right: We use intra-video selection method (VAS) to select summaries from selected unlabeled videos for annotation. We label selected clips by labeling middle frame of each clip. New partially labeled videos are added to the labeled pool for the next iteration.

? Using labeled clips $X(t)$

At iteration t , we train a TAS network and learn action prototypes. More specifically, for each action a in labeled video clips, we learn a prototype p_a . We use the learned TAS model and prototypes to construct a sequence of action prototypes for each partially labeled video. We use the prototype sequences to efficiently select informative unlabeled videos for annotation via our inter-video selection module. The use of prototype sequences instead of entire videos significantly reduces the complexity of video selection.

? Given the trained TAS model at iteration t , we select an informative subset of unlabeled videos from $X(t)$

for partial annotation. To do so, we use a method based on video-prototype sequence alignment (VPA) to select the most useful subset of unlabeled videos. Our key idea is that an unlabeled video whose sequence of actions is sufficiently different from the sequences of actions in partially labeled videos, is expected to improve the TAS model the most, once annotated and used for training. Therefore, we use the prototypes learned by our method to construct action prototype sequences for labeled videos and measure the alignment cost between each unlabeled video and each training prototype sequence. We will select the unlabeled videos with the highest alignment cost for annotation.

? For each selected unlabeled video, we find an informative subset of clips to annotate and to build the labeled clips for the next iteration $X(t+1)$

. To do so, we develop an alignment-based video summarization framework, which finds the

summary of a video that best sequentially aligns with the video. We refer to our method as video-aligned summarization (VAS). This is different from existing summarization methods [91,129] that do not take into account the sequential nature of the summary with respect to the video. Given the NP-hardness of

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the problem, we propose an efficient greedy algorithm to find the subset of informative clips. We use the label of the middle frame in each selected clip as the clip label and obtain $X(t+1)$ by adding the new annotated clips to $X(t)$.

We next discuss each component in more details.

3.1 Learning Action Prototypes and TAS Model

Using annotated video clips at iteration t of active learning, we jointly learn the parameters of a TAS model and action prototypes $\{p_a\}$, representing all seen action classes. These prototypes not only facilitate TAS learning, but also allow constructing compact sequences, which we will use later for efficiently selecting unlabeled videos for annotation.

Regularized Contrastive Learning of Action Prototypes. Our action prototypes are learnable parameters that lie in the embedding space of TAS, each representing an action. Prototypes are learned with features z obtained from the model before the last layer. To learn them, we propose a loss, which we refer to as regularized contrastive action prototype loss, $L_{\text{reg-cont}}$, which consists of three terms,

$$L_{\text{reg-cont}} = L_{\text{cont}} + L_{\mu} + L_{\tau}. \quad (1)$$

Here, L_{cont} is a contrastive loss [15] which enforces that each prototype p_a must be close to labeled clips belonging to action a (positive clips, P_a) and far from labeled clips of other actions (negative clips, N_a),

$$L_{\text{cont}} =$$

$$\sum_a \sum_{x \in P_a} \log \frac{\exp(f(p_a, z_j))}{\sum_{j \in P_a} \exp(f(p_a, z_j)) + \sum_{j \in N_a} \exp(f(p_a, z_j))} \quad (2)$$

In the above, f is the exponential cosine similarity function with a temperature parameter, $f(x, y) = \exp(\cos(x, y)/\tau)$. The contrastive loss enforces that features from the same action to be mapped closely to the associated prototype hence to each other, while having sufficient separation to features of different actions. This leads to obtaining more discriminative action features. Beyond L_{cont} , since active learning adds labeled clips incrementally, the TAS model may encounter different prototypes at various active learning iterations. This can lead to overfitting for some prototypes while underfitting for others. Therefore, we enforce that different prototypes maintain approximately equal levels of separation throughout the learning process. This leads to the following regularization losses

$$L_{\mu} = \frac{1}{M} \sum_a \sum_{x \in P_a} \cos(p_a, p_{a'})$$

$$L_{\tau} = \frac{1}{M} \sum_a \sum_{x \in P_a} \frac{\exp(f(p_a, p_{a'}))}{\sum_{a' \neq a} \exp(f(p_a, p_{a'}))} \quad (2)$$

, (3)
 where $M = A(A \neq 1)$ is the total number of prototype pairs. Experimentally,
 we observed that using these two regularization losses stabilizes the learning
 of prototypes and improves the performance. TAS Learning. To learn the

6. assistant (2026-06-16T03:19:42.628Z)

I have the core of the paper. Let me get the experimental results section with the quantitative numbers.

7. user (2026-06-16T03:19:47.563Z)

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0.2% 0.23% 0.27% 0.35% 0.39% 0.47%
57.5
60.0
62.5
65.0
67.5
70.0
72.5
75.0
90%
95%
100%
f1@10
0.2% 0.23% 0.27% 0.35% 0.39% 0.47%
35
40
45
50
55
90%
95%
100%
f1@50
ours
split-rand
equidistant
split-entropy
coreset
(a)
0.2% 0.23% 0.27% 0.31% 0.35% 0.39%
66
68
70
72
74
90%
95%
f1@10
0.2% 0.23% 0.27% 0.31% 0.35% 0.39%
46
48
50
52
54
56
90%
95%
f1@50
w/o pseudo-label
w/ pseudo-label (b)
Fig. 3: (a) Continue AL iteration up to total 0.47% annotated frames, each horizon-
```


tal line represents the percentage of fully-supervised performance. (b) Continue AL iteration with pseudo labels up to total 0.39% annotated frames, each horizontal line represents the percentage of fully-supervised performance.

4.4 Main Results

Comparison with baselines. Tab. 1 shows comparison of our method with four baselines on 50Salads, GTEA, Breakfast and CrossTask datasets. We fix the budget by running all methods for 4 active learning iterations, equating to total 0.16% frame budget (0.5% for GTEA). We also report the upper bound performance (full), where we use the annotations of all clips during training. Notice on all datasets, the first three baselines have higher performance than coreset. We argue this is mainly because the first three baselines enforce a temporal separation for labeling, indicating that diverse temporal information is crucial for TAS learning. Secondly, we noticed coreset performs relatively better than equidistant on Breakfast and CrossTask, this is partly due to CrossTask and Breakfast being much larger than 50Salads and GTEA, and the lack of intra-video temporal information is compensated by labeling more videos. Overall, our method performs the best across all these datasets, showing the effectiveness of our approach. More specifically, we improve $f1@50$ by 8.2%, 1.8%, 4.1% and 1.2% on 50Salads, GTEA, Breakfast and CrossTask, respectively.

Moreover, by labeling only 0.16% frames (0.5% for GTEA), we achieve 57.8%, 69.1%, 80.1%, and 77.2% of fully-supervised performance on edit distance. Qualitative comparison. Our qualitative analysis, as shown in Fig. 1, compares our method with the split-random baseline across four active learning iterations. A significant difference is evident when examining the segmentation quality of our model, especially when increasing the budget from 0.12% to 0.16%. Compared to the initially trained model and the split-random approach, our model's segmentation quality is substantially better, closely aligning with the ground truth and successfully segmenting all major components. In contrast, the split-random approach fails to identify at least one large segment (the blue one in the middle right), highlighting the superiority of our approach.

Towards Full-supervision Performance. In Fig. 3a, we continue active learning iterations aiming to achieve full-supervision performance on Breakfast. Our method surpasses all baselines over iterations. In particular, with 0.23% frames

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method budget acc edit $f1@\{10,25,50\}$

D-TSTAS (Timestamp) [69] 0.29% 65.7 75.8 71.3 69.3 50.7

EM-gen (SkipTag) [85] 0.29% 64.1 59.9 n/a 57.3 45.2

Ours 0.23% 73.4 71.1 74.4 70.1 54.3

Table 2: Comparison with timestamp and SkipTag supervision methods on Breakfast.

Breakfast MoF IoU IoD CrossTask MoF IoU IoD

TASL [73] 47.8 35.2 46.1 TASL [73] 40.7 14.5 25.1

POC [74] 45.7 38.3 - POC [74] 42.8 15.6 -

MuCon [106] 48.5 40.9 54 MuCon [106] - - -

Ours 63.5 39.2 49.6 Ours 73.9 23.7 30.1

Table 3: Comparison with action set and transcript supervision methods.

labeled, our method achieves over 90% of the full-supervision performance in terms of $f1@10$, while other baselines require 0.27% labeled frames to achieve the same level of performance. Moreover, by labeling 0.35% and 0.47% frames, we attain 95% and 100% of the full-supervision performance, demonstrating the efficiency and effectiveness of our active learning method.

AL in Semi-supervised Settings. One might wonder if trained TAS model can generate pseudo-labels to assist learning in future AL iterations, and we show our method can extend to semi-supervised learning in Fig. 3b. Starting from the 5th iteration, in addition to newly annotated videos, we select 1.25% of videos with the lowest alignment costs and pseudo-label them. We add these pseudo labeled videos to labeled video set for future training. Comparison between learning "w/ pseudo-label" and standard "w/o pseudo-label" setting (orange v.s. blue) shows performance is improved with additional pseudo-labels. More specifically, under 0.31% labeled frames budget, learning with additional pseudo labels achieves close to 95% of fully-supervised performance on $f1@50$, while learning with labeled frames only achieves 90% of fully-supervised performance on $f1@50$. We further notice after 0.35% frames budget, our standard method outperforms semi-supervised one, likely due to noises introduced by pseudo-labels over iterations, resulting in slightly lower performance.

4.5 Comparison with Weakly Supervised Methods

Comparison with timestamp supervision methods. Timestamp supervision [8,50,67,69,85] methods reduce annotation cost by labeling one frame from each action in every video. Examples are D-TSTAS [69] and EM-Gen [85]. Additionally, EM-Gen [85] proposed a new form of supervision named SkipTag, allowing an annotator to randomly label K frames anywhere in the video, where K is the average number of action segments. Although SkipTag reduces the restrictions in timestamp supervision, it still requires all videos to be labeled. Unlike these methods, we do not require watching every entire video or ensuring each

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0.04% 0.08% 0.12% 0.16%

0

10

20

30

40

50

60

12.4

31.9

50.7

57.8

13

24.5

41.8

45.4

acc on 50Salads

0.04% 0.08% 0.12% 0.16%

0

10

20

30

40

5.7

20.5

39.1

45

12.1

22.1

33.5

39.4

edit on 50Salads

w/ VAS

w/o VAS

(a)

0.04% 0.08% 0.12% 0.16%

0

10

20

30

40

50

60

12.4

31.9

50.7

57.8

13.2

29.2

42.7 44.7

acc on 50salads

0.04% 0.08% 0.12% 0.16%

0

10

20
30
40
5.7
20.5
39.1
45
5.7
21.8
32.8
35.2

edit on 50salads

w/ VPA

w/o VPA. (b)

Fig. 4: (a) Ablation of our intra-video selection method (VAS) under 0.04%, 0.08%, 0.12% and 0.16% frame budget on 50Salads. (b) Ablation studies of our inter-video selection method (VPA) under 0.04%, 0.08%, 0.12% and 0.16% frame budget on 50Salads.

selected frame represents a different action segment, significantly simplifying the annotation process while striving for high efficiency and minimal budget.

In Tab. 2, we compare our approach against both D-TSTAS (Timestamp) and EM-Gen (SkipTag) on Breakfast. We adjust γ so we label K frames per selected video while keeping $m = 5\%$ for inter-video selection. Note that our approach annotates only a subset of videos during each iteration, while D-TSTAS and EM-Gen annotate all videos. Consequently, the annotation budget of our method is less than that of D-TSTAS and EM-Gen, and aligns with them only when all videos have been chosen for labeling. Breakfast contains roughly 3.6M frames [67] and $K=6$ [85], which leads to $(1712 \times 6) / 3.6M$ frames $\approx 0.29\%$ budget for D-TSTAS and EM-Gen. For comparison with EM-Gen and D-TSTAS, when selecting 80% videos (0.23% frames), our method significantly outperforms both methods, e.g., surpassing EM-Gen by 9.1% and D-TSTAS by 3.6% on $f1@50$.

In summary, our method performs better than both Timestamp and SkipTag supervision methods, while using less annotation budget.

Comparison with action set and transcript supervision methods. Action set and transcript supervision methods reduce annotation cost by learning from a list of unordered actions or ordered actions, respectively [17]. Quantitatively comparing supervision costs is challenging because obtaining transcripts or action sets typically requires an annotator to watch the entire video. Notably, [67] states that timestamp "does not require more time than annotating transcripts," whereas our supervision cost is lower than timestamp.

We train our method for four AL iterations on Breakfast and CrossTask datasets and compare with transcript methods (TASL [73], MuCon [106]) and an action set method (POC [74]). Tab. 3 demonstrates that our method achieves superior or comparable results while labeling only 0.16% of frames.

4.6 Ablation Studies

Action prototype learning module effectiveness. We evaluate the effectiveness of our action prototype learning module in Tab. 4a and Tab. 4b. For the first row, we use the average embedding features of each action to replace the

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Lcont L μ L? acc edit f1@{10,25,50}

$\times \times \times$ 50.0 44.4 52.2 46.4 26.8

? $\times \times$ 47.2 40.4 45.5 39.7 22.2

? ? $\times$ 42.0 37.8 40.2 33.6 17.8

? $\times$? 49.9 40.6 48.8 42.0 23.9

? ? ? 57.8 45.0 55.1 49.1 32.9

(a) 50Salads

Lcont L μ L? acc edit f1@{10,25,50}

$\times \times \times$ 46.3 56.6 58.9 48.7 27.2

? $\times \times$ 43.9 57.3 57.4 48.0 23.6

? ? $\times$ 46.5 54.9 57.9 48.6 26.4

? $\times$? 42.7 56.6 57.9 47.1 25.9

? ? ? 47.6 57.0 59.9 48.7 27.3

(b) GTEA

Table 4: Loss ablation after 4 active learning iterations, all cases use our AL method. prototypes for selection, and only keep cross entropy loss and smoothing loss for

training. For other cases, we ablate each part in our proposed loss. We run our method for four iterations. Due to lack of labels, using the contrastive loss alone reduces the performance. In contrast, after adding our two regularizers (L_{μ} and L_{γ}), the performance is significantly improved.

Intra-video selection effectiveness. Next, we use split-random to replace our intra-video selection method (VAS), while keeping the inter-video selection (VPA) the same. Figure 4a shows accuracy and edit distance on 50Salads. Notice that VAS outperforms split-random in almost all cases, suggesting the effectiveness of our VAS, creating synergies with our inter-video selection.

Inter-video selection effectiveness. Here, we replace our inter-video selection (VPA) by random selection, while keeping the intra-video selection (VAS). As Figure 4b shows, our method starts from almost the same performance as random (since we also use random for initialization). With more budgets, our method performs better than random selection, suggesting that our proposed inter-video selection (VPA) can better capture useful videos for annotation by selecting action-wise diverse videos. Please refer to the supplementary material for additional experiments, discussions.

5 Limitation and Future Work

Our method is efficient, but the labeled dataset grows iteratively. Future research could explore continual active learning, maintaining a constant-size dataset by cycling videos, potentially reducing training time.

6 Conclusions

We proposed a two-stage active learning framework for TAS, consisting of three parts: i) Regularized action prototype module to learn discriminative prototypes for enhanced accuracy and computational efficiency. ii) Inter-video selection to select unlabeled videos with diverse sequence of actions. iii) Intra-video selection to identify informative clips in selected unlabeled videos for annotation. With extensive experiments on four datasets, we showed our approach consistently outperforms baselines over iterations, and achieve comparable or better performance than other weakly-supervised methods using minimum annotation.

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8. assistant (2026-06-16T03:19:56.379Z)

I now have comprehensive coverage of the paper, including method, quantitative results, ablations, and comparisons. This is a primary research source (ECCV 2024, peer-reviewed, Northeastern University). Let me compile the falsifiable claims.

Key findings relevant to the research question (small-data TAL, active learning, over-segmentation):

- Two-stage AL: inter-video selection (VPA, diversity via DTW alignment) + intra-video selection (VAS, alignment-based summary). First work on AL for TAS.
- Quantitative: labeling only 0.16% frames (0.5% GTEA) achieves 57.8/69.1/80.1/77.2% of fully-supervised edit-distance perf on 4 datasets. f1@50 improved 8.2/1.8/4.1/1.2% over best baseline.
- Towards full supervision: 0.23% frames ? >90% of full-supervision f1@10; 0.35% ? 95%; 0.47% ? 100%. Diversity (temporal separation) matters more than coreset.
- Beats timestamp/SkipTag supervision (D-TSTAS, EM-Gen) at lower budget.
- Semi-supervised extension with pseudo-labels: +pseudo-labels gets to ~95% of full-sup f1@50 at 0.31% budget vs 90% without, but degrades after 0.35% due to pseudo-label noise.

Note: license is not stated in paper; datasets are standard CV benchmarks (cooking/instructional, not sports). I'll flag relevance/transfer caveats.

9. user (2026-06-16T03:20:09.962Z)

Structured output provided successfully

10. assistant (2026-06-16T03:20:24.272Z)

Extracted and verified the source. Summary of what I found:

****Source:**** Su & Elhamifar, "Two-Stage Active Learning for Efficient Temporal Action Segmentation," ECCV 2024 (Northeastern University). Primary peer-reviewed research; full PDF parsed locally (23 pages, all method/results/ablation/reference sections read directly).

****Relevance to the research question (Dimension 3 ? small-data strategies):**** Strongly on-point for the active-learning + semi-supervised-pseudo-label question. The three central claims give concrete, falsifiable numbers on how much fully-supervised quality is recoverable from a tiny labeled fraction, how diversity-driven selection beats core-set, and where pseudo-labeling helps vs. hurts.

****Load-bearing caveats to flag in the synthesis:****

- ****License unverifiable from the paper.**** No code/weights license is stated in the PDF; the hard MIT/Apache/BSD constraint cannot be confirmed from this source alone.
- ****Domain transfer gap.**** All four benchmarks (50Salads, GTEA, Breakfast, CrossTask) are cooking/instructional egocentric video, not sports and not a 1-D shuttle trajectory. The method relies on learned per-action prototypes over a fixed multi-class action dictionary; the indexer's problem is binary rally/dead-time over a low-dimensional trajectory signal, so the inter-video DTW prototype-alignment machinery may not transfer cleanly.
- ****Budget framing differs.**** Their "0.16% of frames" is a fraction of millions of frames across hundreds of videos; the indexer has ~6?16 videos, so the actionable takeaway is the **qualitative** lesson (diversity/temporal-separation selection > random/coreset; pseudo-labels help only at the lowest budgets then degrade from noise) rather than the exact percentages.
- The pseudo-label degradation finding (noise accumulation past 0.35%) directly supports the report's label-noise-handling and "eval?serve"/regression-gate caution.

Saved PDF: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579968447-hsu6x9.pdf`